

Electrical conductivity, induced polarization, and permeability of the Fontainebleau sandstone

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ABSTRACT

The electrical conductivity of clay-free sandstones is customarily assumed to have negligible surface conductivity contribution. The Fontainebleau sandstone, a clean sandstone with relatively coarse ($\sim 250\ \mu\text{m}$) and well-rounded silica grains and silica cement, exhibits surface conductivity along the electrical double layer coating the surface of the grains. A recently developed volume-averaging model for the electrical conductivity was used to determine intrinsic formation factor and surface conductivity from electrical conductivity measurements performed at seven salinities with NaCl solutions. The bulk tortuosity of the pore space influenced the surface conductivity in a predictable way. Formation factor and permeability can be determined as a function of the porosity using the equations developed by Archie for the formation factor, and Revil and

Cathles for the permeability. In both the cases, the data emphasize the existence of a percolation threshold of about 0.02 (2%) in porosity. Once corrected for the effect of this percolation threshold, the porosity exponent of Archie's equation was approximately equal to 1.5 as predicted from the differential effective medium theory for a pack of spherical grains suspended in an electrolyte. We illustrated that permeability can be predicted, within one order of magnitude, from surface conductivity, porosity, and formation factor. Spectral-induced polarization data indicated that the in-phase conductivity was nearly frequency-independent (in the frequency range from 1 to 10 kHz) whereas the quadrature conductivity displayed a relationship between the surface conductivity and a peak frequency likely related to the pore throat size determined from mercury porosimetry measurements.

INTRODUCTION

The Fontainebleau sandstone is a clean sandstone of Oligocene age outcropping in the Paris Basin (France). It was deposited during the Stampian in dunes bordering the shore of the Paris Basin. Grains are monocrystalline ($\sim 99.8\%$ silica according to X-ray diffraction (XRD) Kieffer et al., 1999), well-rounded (surface roughness close to 1; see Adams et al. [2012], for some definitions of the surface roughness and Kieffer et al. [1999] for an application to a discussion on the Fontainebleau sandstone), and well-sorted (Bourbié et al., 1987; Kieffer et al., 1999). The grain-size distribution is relatively homogeneous with a mean grain diameter around $250\ \mu\text{m}$ and grain

sizes typically in the range $150\text{--}300\ \mu\text{m}$ (Bourbié and Zinzner, 1985; Doyen, 1988).

The porosity of Fontainebleau sandstone is intergranular and covers a continuous range between 0.03 and 0.30 without noticeable grain-size modification (Bourbié and Zinzner, 1985). This porosity range is due to cementation by silica precipitation (Alimen, 1936; Thiry et al., 1988). A high-porosity end-member would correspond to the Fontainebleau sand with a porosity in the range 0.38–0.50 depending on its state of consolidation. Due to this broad porosity range and its relatively narrow grain-size distribution, the Fontainebleau sandstone has been widely used to infer and/or test permeability/porosity relationships (Bourbié and Zinsner, 1985; Bourbié

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et al., 1987; du Plessis and Roos, 1994; Mavko and Nur, 1997; Kieffer et al., 1999) and electrical conductivity models for saturated and unsaturated conditions (Durand, 2003; Knackstedt et al., 2007; Han et al., 2009; Gomez et al., 2010; Yanici et al., 2013). The advantage offered by the Fontainebleau sandstone is to study the effects of porosity on transport properties independently of every other parameter such as the grain diameter and coordination number between pores or the effect of detrital or authigenic clay minerals.

The relationship between permeability and electrical conductivity has not been explored in a satisfactory manner despite the fact that the inference of such a relationship has many practical and fundamental applications in hydrogeology, hydrogeophysics, oil reservoir exploration and production, and civil engineering. Indeed, the effect of surface conductivity has not been accounted for in all previous published works (with one exception Börner [1992], for a single sample) whereby the apparent formation factor (defined as the ratio of the pore fluid to sample electrical conductivity) can be different from the intrinsic formation factor (corrected for surface conductivity) by more than an order of magnitude at low salinities (0.001 S m^{-1}). We therefore believe that this problem needs to be addressed in more details for the following reasons:

- 1) We need to develop fundamental methods for the interpretation of electrical resistivity tomography (ERT) and time-lapse ERT data under field conditions (Kemna, 2000; Kim et al., 2009; Johnson et al., 2010; Karaoulis et al., 2011). With few exceptions (e.g., Müller et al., 2010; Jardani et al., 2013; Revil et al., 2013a, 2013b), most of the published works in hydrogeophysics have neglected the contribution of surface conductivity occurring along the surface of the grains in the electrical double layer (for instance, Kowalsky et al., 2011; Pollock and Cipka, 2012). Few works have considered the existence of surface conductivity for clay-free materials. Exceptions include the works of Rink and Schopper (1974), Börner (1992), and Lorne et al. (1999). The effect of neglected surface conductivity is uncertain in freshwater aquifers or even in hydrocarbon-bearing rocks. Additionally, the independent estimation of surface conductivity needed to separate bulk from surface conductivity responses in the field has been challenging.
- 2) The recent model of Revil (2013a, 2013b) appears to explain a broad number of complex electrical conductivity data in saturated and unsaturated conditions including the case of clayey porous media. Complex conductivity involves, in addition to an in-phase conductivity component (the real component or simply called the *electrical conductivity* below), a quadrature (imaginary) component associated with the reversible storage of electrical charged in an alternating electrical field. The model of Revil (2013a, 2013b) is based on the volume-average of the local Nernst-Planck equation up to a representative elementary volume of a porous rock. It appears to reliably predict the salinity, tortuosity, and saturation dependence of the complex conductivity data of clayey porous rocks. That said, this model still needs to be further examined, validated, or falsified. The Fontainebleau sandstone offers a reliable way to test some of the fundamental predictions of this model regarding, for instance, the dependence of surface and quadrature conductivities with bulk tortuosity of the pore space for a clean sandstone, the difference between apparent and intrinsic formation factors, and the relationship between surface and quadrature conductivities.

- 3) The holy grail in hydrogeophysics is to image permeability using geophysical data or a combination of in situ measurements (e.g., pumping tests) and geophysical data (e.g., Kowalsky et al., 2004; Linde et al., 2006; Costar et al., 2008). As ERT and induced polarization are increasingly popular in hydrogeophysics to reach this goal (e.g., Kemna, 2000), it is interesting to verify what can be achieved on a relatively simple material like the Fontainebleau sandstone. We explore whether permeability can be inferred from surface conductivity, formation factor, and porosity and if affirmative, with what uncertainty.
- 4) There is a need to explain regions of high electrical conductivity (10^{-3} to 10^{-1} S m^{-1}) in the crust observed through magnetotelluric data and the role of quartzite in these observations (see for instance Shimojuku et al. [2014] and references therein). It seems that the effect of surface conductivity has been totally ignored in these studies.

In this paper, we first test the electrical conductivity model developed by Revil (2013a) on the Fontainebleau sandstone. We show that the electrical conductivity of a clay-free sandstone such as the Fontainebleau sandstone has a nonnegligible contribution of surface electrical conductivity (associated with electromigration in the electrical double layer coating the surface of the grains), but that surface conductivity can be quantified in terms of a simple petrophysical model. We also show that the formation factor and the permeability can be expressed as a function of a reduced porosity (connected porosity minus a percolation porosity of 0.02 [2%]) using the equations developed by Archie (1942) and Revil and Cathles (1999), respectively. We will also discuss the validity of the second Archie's equation for Fontainebleau sandstone, and finally, we document preliminary data for spectral-induced polarization phenomena.

ELECTRICAL CONDUCTIVITY AND PERMEABILITY MODELS

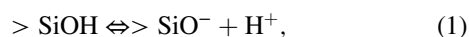
Surface conductivity in silica sands

A popular belief, in the petroleum engineering and hydrogeophysical communities, is to consider silica sands with no clays ("clean sands") as having only a bulk electrical conductivity component associated with the electromigration of the charge carriers in the pore space of the material. This belief, however, does not agree with the observations. Clay-free silica sands are characterized by surface conductivity (e.g., Rink and Schopper, 1974; Börner, 1992), which occurs along the surface of their grains along the electrical double layer (Revil and Glover, 1997; Revil et al., 1999). The existence of this conduction mechanism for silica has been known for a long time in other communities (e.g., in colloidal chemistry; see Watillon and de Backer, 1970) but ignored or dismissed in applied geophysics for "clean" sandstones. Although in some cases, such dismissal can be proved correct, it is rarely verified whether the surface conductivity of clay-free sandstones and sandy aquifers/reservoirs is negligible or not in freshwater environments.

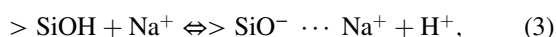
The surface of silica grains is coated by an electrical double layer, which is sketched in Figure 1. This double layer comprises a Stern layer of weakly sorbed counterions (Stern, 1924) plus a diffuse layer of counterions and co-ions (Gouy, 1910; Chapman, 1913). The counterions and co-ions of the diffuse layer interact with the charged mineral surface only through Coulomb-type interactions. Electromigration in the electrical double layer of silica particles

or grains is responsible for surface conductivity (see Leroy et al., 2013 for an explicit model of surface conductivity along silica grains and accounting for the effect of complexation of the active sites on the surface of silica grains).

The charge on the mineral surface is due to chemical reactions between the pore water and the mineral surface responsible for the presence of charged surface sites. For instance, the speciation of the silanol surface sites ($> \text{Si-OH}$, “ $>$ ” denotes the crystalline framework) on the silica surface in contact with a dissociated salt like Na^+Cl^- can be described by the following speciation model (Davis et al., 1978):



and



where the symbol “ $\cdots$ ” is used to explain that the sorption of the counterions does not involve a covalent bound. Reaction 3 corresponds to the (weak) sorption of the sodium in the Stern layer (see the discussion in Revil, 2012, 2013a). This sorption is usually weak as the sorbed counterions retain their hydration shell and are possibly characterized by a mobility in an electrical field that is the same as in the bulk pore water and enables a surface conductivity (Revil, 2013b). The situation is different for clay minerals as discussed by Revil (2012). In clay materials, it seems that the mobility of the counterions in the Stern layer is much lower than in the diffuse layer and in the bulk pore water (Revil, 2012; Weller et al., 2013).

Electrical conductivity model

When surface conductivity is small with respect to pore water conductivity, the conductivity of a porous material σ can be linearly related to the conductivity of pore water σ_w by a Waxman and Smits (1968)-type relationship. This relationship linearly connects the conductivity of the material σ and the conductivity along the surface of the grains (surface conductivity σ_s) (e.g., Weller et al., 2013) as follows:

$$\sigma = \frac{1}{F} \sigma_w + \sigma_s. \quad (4)$$

Surface conductivity in equation 4 is defined in a macroscopic sense as the conductivity of a representative elementary volume associated with local conduction in the electrical double layer at the local scale around the grains. In equation 4, F denotes the (intrinsic) formation factor, which is a characteristic of the pore network. This formation factor is related to the connected porosity by the Archie's equation:

$$F = \phi^{-m}, \quad (5)$$

where m denotes the porosity exponent (sometimes called the *cementation exponent*). In clayey materials, Revil (2013a) obtain an equation that is similar to equation 4 in the so-called high-salinity

limit (dictated by the dominance of the distribution of the bulk conductances) but with some important modifications with respect to the Waxman and Smits' model. In the model developed by Revil (2013a) and for frequencies above the frequency of polarization of the grains (typically > 1 Hz), the surface conductivity is related to the cation exchange capacity (CEC, in C kg^{-1}) of the porous material by

$$\sigma_s = \Theta \left(\frac{1}{F\phi} \right) \rho_s [\beta_{(+)}(1-f) + \beta_{(+)}^s f] \text{CEC}, \quad (6)$$

where $\beta_{(+)}^s$ denotes the mobility of the counterions in the Stern layer (in $\text{m}^2 \text{V}^{-1} \text{s}^{-1}$), $\beta_{(+)}$ represents the mobility of the cation in the pore water (in $\text{m}^2 \text{V}^{-1} \text{s}^{-1}$), ρ_s represents the mass density of the grains (typically 2650 kg m^{-3} for silica), f quantifies the fraction of counterions in the Stern layer with respect to the total concentration of cations in the electrical double layer, and the correction factor Θ is given by

$$\Theta = \left(\frac{F-1}{F} \right) \approx 1. \quad (7)$$

Indeed, in our case, the smallest formation factor will be around 10, which leads to $\Theta \approx 0.90$. For silica, Revil (2013a, 2013b) finds that $\beta_{(+)}^s \approx \beta_{(+)}$ and $\beta_{(+)}(\text{Na}^+, 25^\circ\text{C}) = 5.2 \times 10^{-8} \text{ m}^2 \text{V}^{-1} \text{s}^{-1}$. For silica grains, Revil (2013b) obtains a relationship between the mean grain diameter and the equivalent CEC of the material

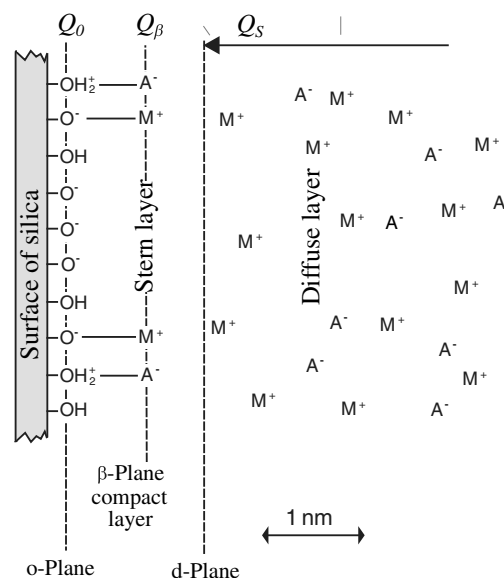


Figure 1. Sketch of the electrical double layer coating the surface of the grains. The symbol M represents the metal cations (e.g., Na^+ or K^+) and A the anions (e.g., Cl^-). The charge density Q_0 (in C m^{-2}) denotes the surface charge density of the surface of the mineral (located on the o -plane), whereas Q_β (in C m^{-2}) is the surface charge density of the Stern layer (compact layer located on the β -plane), and Q_s (in C m^{-2}) denotes the equivalent surface charge density of the diffuse layer (the sum of the three charge densities is equal to zero). The d -plane separates the outer part of the Stern layer from the diffuse layer. The o -plane represents the true mineral surface whereas the β -plane represents the mean plane for the sorption of the counterions in the Stern layer.

given by $\text{CEC} = 6 Q_S/(\rho_s d)$. For well-rounded grains with $Q_S = 0.64 \text{ C m}^{-2}$ (equivalent charge density on the mineral surface), $d = 250 \text{ }\mu\text{m}$, and $\rho_s = 2650 \text{ kg m}^{-3}$, we obtain a CEC of 5.80 C kg^{-3} . This equivalent CEC represents the amount of exchangeable cations located on the surface of the mineral at a given pore water composition including the effect of pH. Consequently, the surface conductivity can be determined from

$$\sigma_s = 6Q_S\beta_{(+)} \frac{1}{d} \left(\frac{1}{F\phi} \right). \quad (8)$$

Revil and Florsch (2010) introduce a method to extend equation 8 to the case of a grain-size distribution characterized by a probability density function. However, this extension is not required here because of the quite narrow grain-size distribution of the Fontainebleau sandstone (see discussion below in the “Induced Polarization” section).

The macroscopic effect of surface conductivity can be readily observed by having the grains coated with a thin layer of equivalent conductivity $2x_d Q_S \beta_{(+)}$, where x_d corresponds to the Debye screening length ($2x_d$ is roughly a measure of the thickness of the electrical double layer; see Strauss, 1954; Revil and Glover, 1997). Using $Q_S = 0.64 \text{ C m}^{-2}$, $x_d = 2 \text{ nm}$, and $\beta_{(+)}(\text{Na}^+, 25^\circ\text{C}) = 5.2 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ yields a conductivity of 0.7 S m^{-1} for this layer. Actually because of the effect of increasing sorption of counterions with the salinity, the conductivity of this layer would probably increase slightly with the salinity but this increase will be neglected below.

Temperature would mainly impact the mobilities of the ions in the expression of the conductivity of the pore water and the mobility of the counterions in the diffuse layer. These mobilities are related to the viscosity of the pore water and therefore are typically expected to change by approximately 2% per degree Celsius (see the discussion in Revil [2012], section 3.5).

According to equation 8, the macroscopic surface conductivity σ_s is also inversely proportional to the bulk (electrical) tortuosity α (≥ 1) of the pore space defined by (e.g., Pride, 1994)

$$\alpha = F\phi. \quad (9)$$

Note that sometimes in hydrogeology, tortuosity is defined as the reciprocal of α (see for instance Bear, 1988; Bear and Bachmat, 1991). Equation 8 implies that for a constant tortuosity of the pore space (for instance for sand packs with all the same porosity), the surface conductivity is inversely proportional to grain diameter. This prediction was actually documented by Bolève et al. (2007, their Figure 6) for silica sands at the same porosity. The other extreme case, untested so far, concerns a material characterized by the same grain diameter but having different tortuosities. In this situation, the surface conductivity is assumed to be inversely proportional to the bulk tortuosity of the material. Therefore, in the case of the Fontainebleau sandstone, we expect to observe an increase in surface conductivity when tortuosity decreases, which corresponds to an increase in porosity in the case of this sandstone. This effect should be predictable according to our model. To the authors' knowledge, such a prediction has never been checked in the open scientific literature.

Another point of interest is the inclusion of a percolation threshold in the relationship between formation factor and porosity. This yields the modified version of Archie's equation proposed by Sen et al. (1981) as

$$F = (\phi - \phi_p)^{-m^*}, \quad (10)$$

where ϕ_p denotes a porosity at the percolation threshold and m^* denotes a modified Archie's exponent used in a differential effective medium approach for which there is no percolation threshold. When they analyzed their data, Sen et al. (1981) realize of the need to introduce such a percolation threshold in the formation factor/porosity relationship. Percolation theory is not of great help in our case because it is only valid in the close vicinity of the percolation threshold (Guéguen and Palciauskas, 1994), i.e., for porosities close to 0.02 (2%) in our case as discussed below. Using percolation concepts, Guéguen and Palciauskas (1994) obtain a formation factor/porosity relationship $F \propto (\phi - \phi_p)^{-2}\phi^{-1}$. Such an approach should rule in the vicinity of the percolation threshold and should possibly be used rather than models based on volume-averaging or differential effective medium schemes.

We also believe that the same percolation threshold applies to the full data set because the reduction in porosity for all the samples obeys the same process, namely the precipitation of a silica cement in a quartz sand (French and Worden, 2013). Numerical investigations show that if the diameter of the grains grows uniformly during the porosity reduction process, the percolation porosity is expected to be in the range $\sim 0.02 - 0.05$ (Roberts and Schwartz, 1985; Schwartz and Kimminau, 1987). Because the Fontainebleau sandstone exhibits a continuous cementation process, it should clearly exhibit a percolation threshold in the formation factor porosity relationship as discussed by Sen et al. (1981) and Revil (2002) among others.

Permeability

According to the model developed by Revil and coworkers (Revil and Cathles, 1999; Revil, 2002; Revil and Florsch, 2010), the permeability k (in m^2) of granular material is given by

$$k = \frac{d^2}{aF(F-1)^2} \approx \frac{d^2}{aF^3}, \quad (11)$$

using $F \gg 1$ and where a is a constant on the order of 8 according to Revil (2002). Note however that Revil and Cathles (1999) and Revil and Florsch (2010) use higher values between 24 and 32. If we combine the approaches of Avellaneda and Torquato (1991) and Revil and Cathles (1999), it appears that such a parameter is dependent on the pore shape. If we replace the formation factor by Archie's equation in equation 11, the permeability exhibits a power law dependence on porosity. Such behavior has been empirically known since Rose (1945) and is first modeled by Revil and Cathles (1999).

Because permeability can be related to the formation factor, the same percolation threshold should apply to the permeability and the formation factor. If we use the modified Archie's equation instead of the classical Archie's equation, the permeability should be related to the reduced porosity $\phi - \phi_p$ by the following power-law relationship:

$$k \approx \frac{d^2}{a} (\phi - \phi_p)^{3m^*}. \quad (12)$$

According to equation 12, if the formation factor exhibits a percolation threshold, we also expect the permeability to exhibit the same percolation threshold. Mavko and Nur (1997) introduce a percolation threshold in the Kozeny-Carman equation to improve the prediction of permeability for tight sandstones. However, the Kozeny-Carman equation overpredicts the permeability of sandstones (Revil and Cathles, 1999). The introduction of a percolation threshold in volcanic rocks and magma mush has also been reported in various studies (e.g., Cheadle et al., 2004).

COMPARISON TO EXPERIMENTAL DATA

Electrical conductivity

The set of core plugs analyzed in this study comprises 69 Fontainebleau sandstones collected at outcrops in the Ile de France region, around Paris, France, and provided by the Institut Français du Pétrole in France. Core plugs have a diameter of 2.0 cm and a length ranging between 1.5 and 4.2 cm. The microstructure of four of them, characterized by distinct porosities, is shown in Figure 2. These images of the dry pore-space were obtained with a 4- μm resolution by a micro-X-ray CT imaging system located at the Colorado School of Mines. The mineralogy of these samples is 99.98% silica (grain and cement) (through XRD analysis), and they are characterized by well-rounded (surface roughness close to 1) and well-sorted grains (Figure 2). Table 1 summarizes the connected porosity ϕ determined from triple weight measurements (Archimedes's method including dry and water-saturated weights and the weight of the same saturated samples immersed in water).

Electrical conductivity measurements were conducted at seven salinities with NaCl solutions. These salinities correspond to the following pore fluid conductivities $\sigma_w = (1.70 \times 10^{-3}, 6.40 \times 10^{-2}, 0.12, 0.49, 0.98, 5.06, \text{ and } 18.7 \text{ S m}^{-1})$ at room temperature ($\sim 20 \pm 2^\circ\text{C}$), therefore covering four orders of magnitude. Water saturation was performed under vacuum (few Pa) to ensure a complete saturation of the core samples and to avoid air entrapment. Once saturated, the samples were immersed in their electrolyte for several weeks prior to acquiring any set of measurements at a given salinity to be certain that chemical equilibrium was reached. The electrical conductivity measurements were performed using a two-electrode system with a 1260 Solartron impedance meter at a frequency of 4 kHz to avoid contamination by the polarization of the electrodes (see Revil et al., 1996). The instrumental error (related to the voltage measurement) for the measured resistivities was 2%.

Figure 3a shows the conductivity data for four samples characterized by distinct porosities, and Figure 3b shows the same trend for Fontainebleau sandstone Core F3 from the database of Börner (1992). These conductivity data confirm a high degree of linearity between the conductivity of the sandstone and the pore water conductivity. Figure 3 shows that surface conductivity is lower at low porosity than at high porosity. Indeed, the samples characterized by the higher porosities are also the ones associated with the highest surface conductivity and vice versa. All the conductivity data were first fitted with equation 4 to determine the formation factor F and the surface conductivity σ_s assuming that the salinity dependence of

the surface conductivity can be neglected (the salinity and pore water composition dependence of surface conductivity are discussed in Vaudelet et al. [2011a, 2011b] and will be neglected in this study as a first-order approximation; e.g., Revil, 2012). The values of F and σ_s are reported in Table 1 for the 69 samples.

In Figure 4, we report the values of formation factor as a function of connected porosity. We use the modified Archie's equation, equation 10, to determine the mean value of the porosity exponent m and the value of the percolation porosity ϕ_p . The best fit of equation 10 yields $m = 1.46 \pm 0.05$ and $\phi_p = 0.019 \pm 0.004$ (optimization is done with a Gauss-Newton algorithm). The fitted value of m (1.46 ± 0.05) is close to the theoretical value $m = 1.5$ according to the differential effective medium theory applied to a pack of spherical grains suspended in a background electrolyte (see Sen et al., 1981). This agrees with the well-rounded nature of the grains. We observe in Figure 4 that the data exhibit significant scattering close to the percolation threshold. This behavior shows the limitation of mean-field theory in the vicinity of the percolation threshold with a representative elementary volume that exceeds the volume of the core samples. That said, the experimental data fluctuate around the predicted values of 1.5 and more precisely between 1.40 and 1.60 for the core samples with a porosity higher than 0.08 (8%).

In Figure 5, we determine the cementation exponent for each sample accounting (Figure 5b) or not (Figure 5a) for the percolation porosity in the Archie's equation. The porosity exponent m entering the classical Archie's equation increases with the cementation of the material and the concomitant decrease in porosity (Figure 5a). When corrected for the effect of percolation threshold, it is clear that the value of the cementation exponent is close to the theoretical

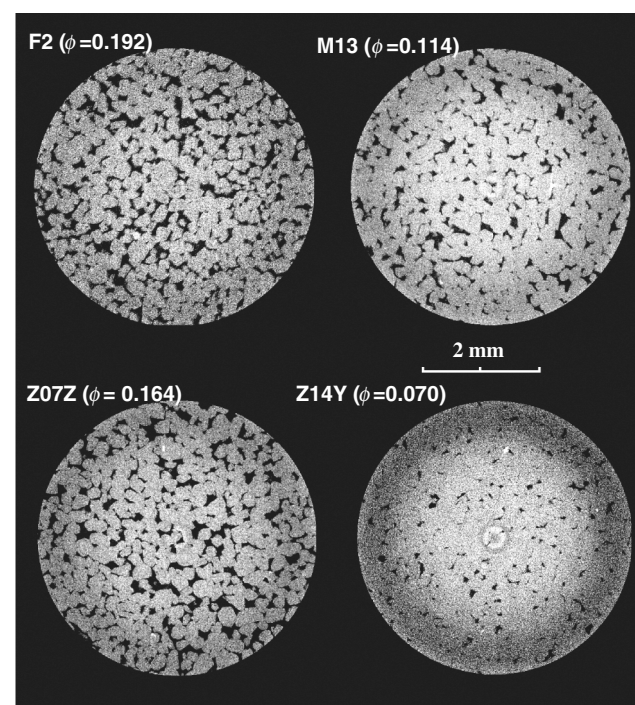


Figure 2. CT scans for the Fontainebleau core samples F2, M13, Z07Z, and Z14Y. The scale, given as a white line, is 2000 μm (note that the grains have a diameter close to 250 μm). The porosity of these core samples is given as void percentage in the figures, and their remaining properties are reported in Table 1.

Table 1. Main experimental results obtained from 69 core samples of the Fontainebleau sandstone: ΔF and $\Delta\sigma_s$ represent twice the standard deviations of formation factor and surface conductivity, respectively. In this table, r represents the correlation coefficient describing the quality of the fit of the electrical conductivity data using equation 4.

Sample	ϕ (fraction)	k (m ²)	F (-)	σ_s (10 ⁻⁴ S m ⁻¹)	ΔF (-)	$\Delta\sigma_s$ (10 ⁻⁴ S m ⁻¹)	r (-)
Z01X	0.066	1.82×10^{-14}	52.38	1.54	1.13	0.09	0.99984
Z01Y	0.054	3.35×10^{-15}	75.39	1.03	1.71	0.07	0.99983
Z01Z	0.057	6.30×10^{-15}	99.56	0.92	1.24	0.03	0.99995
Z02X	0.075	2.11×10^{-14}	49.12	1.57	1.01	0.09	0.99985
Z02Y	0.069	4.97×10^{-14}	39.85	1.78	0.99	0.13	0.99979
Z02Z	0.050	2.49×10^{-15}	84.63	1.10	1.47	0.05	0.99990
Z03Y	0.106	1.54×10^{-13}	38.22	1.63	1.06	0.14	0.99975
Z03Z1	0.102	1.55×10^{-13}	48.14	1.49	1.18	0.11	0.99980
Z03Z2	0.097	1.33×10^{-13}	38.52	1.78	0.69	0.09	0.99989
Z04X	0.068	1.02×10^{-14}	79.65	1.05	0.80	0.03	0.99997
Z04Y1	0.202	3.40×10^{-12}	11.35	4.37	0.44	0.55	0.99950
Z04Y2	0.067	1.21×10^{-14}	103.01	1.03	1.29	0.03	0.99995
Z04Z1	0.222	3.56×10^{-12}	10.10	5.14	0.30	0.49	0.99970
Z04Z2	0.073	1.09×10^{-14}	129.59	1.18	0.89	0.02	0.99998
Z05Y	0.084	4.23×10^{-14}	51.45	1.90	0.99	0.10	0.99987
Z06X	0.078	2.29×10^{-14}	57.23	1.84	1.49	0.13	0.99976
Z06Y	0.085	5.81×10^{-14}	57.66	1.65	0.82	0.06	0.99993
Z06Z	0.085	1.72×10^{-13}	45.87	1.99	1.52	0.18	0.99962
Z07X	0.168	1.64×10^{-12}	15.35	4.04	0.35	0.28	0.99982
Z07Z	0.164	1.69×10^{-12}	15.20	3.35	0.31	0.22	0.99987
Z08X	0.159	1.77×10^{-12}	17.07	3.26	0.63	0.37	0.99955
Z08Y	0.155	1.54×10^{-12}	19.21	4.13	0.47	0.29	0.99979
Z08Z	0.151	1.22×10^{-12}	19.83	3.32	0.60	0.30	0.99969
Z09Y	0.207	3.12×10^{-12}	11.03	4.80	0.36	0.49	0.99966
Z09Z	0.212	4.52×10^{-12}	10.16	5.50	0.36	0.61	0.99959
Z10X	0.174	1.35×10^{-12}	17.06	4.03	0.37	0.26	0.99984
Z10Y	0.170	9.81×10^{-13}	19.22	3.88	0.56	0.33	0.99971
Z10	0.158	7.02×10^{-13}	20.13	3.21	0.49	0.24	0.99980
Z13X1	0.100	1.90×10^{-13}	40.80	2.47	0.78	0.13	0.99987
Z13X2	0.159	1.28×10^{-12}	18.05	4.02	0.51	0.33	0.99974
Z13Y	0.159	1.19×10^{-12}	17.18	3.88	0.41	0.27	0.99981
Z13Z	0.170	1.09×10^{-12}	17.88	2.79	0.56	0.28	0.99969
Z14X	0.045	1.90×10^{-15}	135.52	0.96	4.78	0.09	0.99956
Z14Y	0.070	2.42×10^{-14}	117.31	1.02	3.15	0.07	0.99975
Z15X	0.092	1.82×10^{-13}	27.71	1.98	0.93	0.21	0.99963
Z15Y	0.109	1.49×10^{-13}	32.38	1.77	0.77	0.13	0.99981
Z16X	0.069	1.57×10^{-14}	38.79	1.95	1.03	0.15	0.99976
Z16Z	0.069	2.36×10^{-14}	32.72	2.59	0.73	0.16	0.99983
Z17X1	0.180	1.74×10^{-12}	14.64	2.99	0.49	0.33	0.99964
Z17X2	0.188	2.28×10^{-12}	16.34	4.12	0.54	0.40	0.99962
Z17Y	0.189	2.18×10^{-12}	12.75	2.81	0.48	0.38	0.99955
Z17Z	0.194	3.39×10^{-12}	12.67	2.28	0.36	0.25	0.99975
Z18X	0.052	1.20×10^{-15}	182.50	0.57	5.17	0.04	0.99972
Z18Y	0.047	1.30×10^{-15}	180.86	0.61	2.22	0.02	0.99995
Z18Z	0.059	5.30×10^{-15}	88.18	1.26	1.43	0.05	0.99991

Table 1. Main experimental results obtained from 69 core samples of the Fontainebleau sandstone: ΔF and $\Delta\sigma_s$ represent twice the standard deviations of formation factor and surface conductivity, respectively. In this table, r represents the correlation coefficient describing the quality of the fit of the electrical conductivity data using equation 4. (continued)

Sample	ϕ (fraction)	k (m ²)	F (-)	σ_s (10 ⁻⁴ S m ⁻¹)	ΔF (-)	$\Delta\sigma_s$ (10 ⁻⁴ S m ⁻¹)	r (-)
Z19X	0.108	2.82×10^{-13}	39.96	1.55	0.88	0.10	0.99984
Z19Y	0.103	3.99×10^{-13}	38.56	1.34	0.72	0.08	0.99989
Z19Z	0.099	2.00×10^{-13}	49.05	2.12	1.19	0.14	0.99979
Z20X	0.056	6.10×10^{-15}	145.32	0.71	4.43	0.06	0.99968
Z20Y	0.052	4.70×10^{-15}	141.34	0.84	2.74	0.04	0.99987
M1	0.080	4.10×10^{-14}	71.13	1.85	1.65	0.11	0.99981
M2	0.111	3.24×10^{-13}	37.79	1.55	0.48	0.06	0.99995
M3	0.113	3.59×10^{-13}	34.65	1.48	0.87	0.12	0.99980
M5	0.062	1.02×10^{-14}	228.59	0.77	2.16	0.02	0.99996
M6	0.094	9.92×10^{-14}	52.45	1.57	1.10	0.09	0.99985
M7	0.202	4.08×10^{-12}	11.40	4.91	0.32	0.43	0.99974
M8	0.045	5.24×10^{-16}	342.50	0.41	5.31	0.02	0.99991
M10	0.055	4.10×10^{-15}	307.09	0.50	5.70	0.02	0.99988
M11	0.160	1.43×10^{-12}	17.19	3.38	0.58	0.35	0.99962
M12	0.051	2.17×10^{-15}	289.33	0.46	9.03	0.04	0.99966
M13	0.114	3.16×10^{-13}	31.19	1.75	1.08	0.19	0.99960
M14	0.077	4.44×10^{-14}	92.14	1.36	2.85	0.11	0.99966
M16	0.171	1.27×10^{-12}	17.38	3.70	0.65	0.38	0.99949
M17	0.115	3.50×10^{-13}	33.30	1.58	0.85	0.13	0.99979
F2	0.192	2.88×10^{-12}	12.35	2.91	0.37	0.31	0.99972
F8	0.054	1.91×10^{-15}	221.62	0.51	6.48	0.51	0.99970
F9	0.126	5.64×10^{-13}	32.43	1.68	0.81	0.12	0.99978
F81Y	0.042	3.91×10^{-16}	266.09	0.47	4.01	0.02	0.99992
PHI52	0.071	1.72×10^{-14}	52.18	1.34	1.37	0.09	0.99975

value predicted for a pack of spherical grains immersed in a background electrolyte ($m = 1.5$; see Figure 5b).

Figure 6 shows the bulk tortuosity of the pore space $\alpha = F\phi = (\phi - \phi_p)^{-m}\phi$, determined using equation 9, as a function of connected porosity. The data are fitted with equation 9 in which the formation factor is replaced by equation 10 and using the theoretical value of the porosity exponent ($m = 1.5$). The bulk tortuosity of the pore space increases strongly close to the percolation threshold. At a porosity of 0.05, the bulk tortuosity reaches values in the range from 10 to 20.

Surface conductivity and tortuosity

A crucial test of the model developed by Revil (2013a) is the dependence of conductivity with bulk tortuosity of the pore space. From equation 8, we can write the relationship between surface conductivity and bulk tortuosity of the pore space as

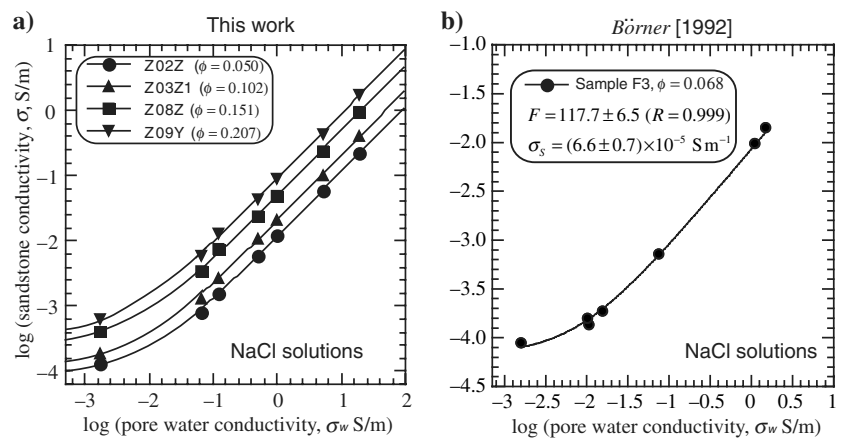


Figure 3. Electrical conductivity of the porous material versus pore fluid electrical conductivity using NaCl solutions. (a) The log-log plot shows the electrical conductivity data of four representative samples (at four distinct porosities from 0.05 to 0.21) and for seven pore water conductivities (NaCl solutions). The lines correspond to the fit with the linear model discussed in the main text and is used to determine the (intrinsic) formation factor and the surface conductivity according to equation 4. The salinity dependence of surface conductivity (e.g., Leroy and Revil, 2004; Weller et al., 2013) is neglected following the same (b) for the sample F3 of Börner (1992).

$$\log \sigma_s = b - \log(F\phi), \quad (13)$$

where b is defined by

$$b \equiv \log\left(\frac{6Q_s\beta_{(+)}}{d}\right). \quad (14)$$

The term b is a constant as long as all the samples have the same mean grain diameter d , which is approximately true for Fontainebleau sandstone. Taking $Q_s = 0.64 \text{ C m}^{-2}$ (see Revil, 2012; 2013a, 2013b), $d = 229 \text{ }\mu\text{m}$ (same as for fitting the permeability–porosity data) below and consistent with the $250 \text{ }\mu\text{m}$ given by Bourbié and Zinzner (1985) and $\beta_{(+)}(\text{Na}^+, 25^\circ\text{C}) = 5.2 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ yields $b = -3.06$. This value is in agreement with the experimental data displayed in Figure 7. Indeed, the best fit of equation 13 yields $b = -3.11 \pm 0.02$ (the optimization is done with a Gauss-Newton algorithm accounting only for the uncertainty related to the surface conductivity data, themselves derived from the fit of the electrical conductivity data). This value is therefore close to the theoretical values given above (-3.06) using equation 14. The data displayed in Figure 7 shows an inverse relationship between surface conductivity and bulk tortuosity as predicted from the model introduced by Revil (2013a) (see equation 8 above).

Permeability

Permeability was determined for the 69 samples using an air permeameter and applying the Klinkenberg correction due to gas slippage (Klinkenberg, 1941). The permeability of the investigated samples (reported in Table 1) is in the range 5×10^{-12} – $4 \times 10^{-16} \text{ m}^2$ for

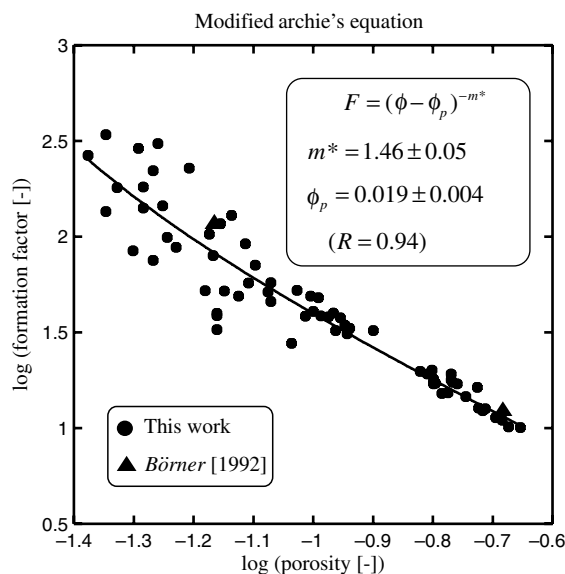


Figure 4. Electrical formation factor versus porosity for the complete set of samples. A modified Archie's relationship is used to determine the log–log relationship between electrical formation factor and porosity introducing a percolation threshold. The percolation porosity is in the range from 0.015 to 0.023 (best fit: 0.019) and the cementation exponent is in the range 1.4–1.5 (best fit: 1.46). The data points from Börner (1992) concern the core samples F1 ($F = 12.5$, $\phi = 0.214$) and F3 ($F = 118$, $\phi = 0.676$).

a porosity ranging from 0.04 to 0.21. Consequently, the range of permeability investigated covers four decades, which is a fairly broad range for a clean sandstone.

Before discussing the prediction of permeability using electrical conductivity data, we discuss the concept of equivalent grain diameter in the presence of cementation. In the case of the Fontainebleau sandstone, one may wonder whether the notion of grain diameter is still valid at low porosities. As shown in Figures 2 and 8, the cementation of grains at low porosities may yield to bigger equivalent grain diameters. This concept is, however, not supported by the data because all the data can be fitted with a single value of the grain diameter.

We start by comparing permeability with the formation factor (Figure 9) and with porosity (Figure 10). The relationship between permeability and connected porosity displays much less scatter than the relationship between permeability and formation factor. This

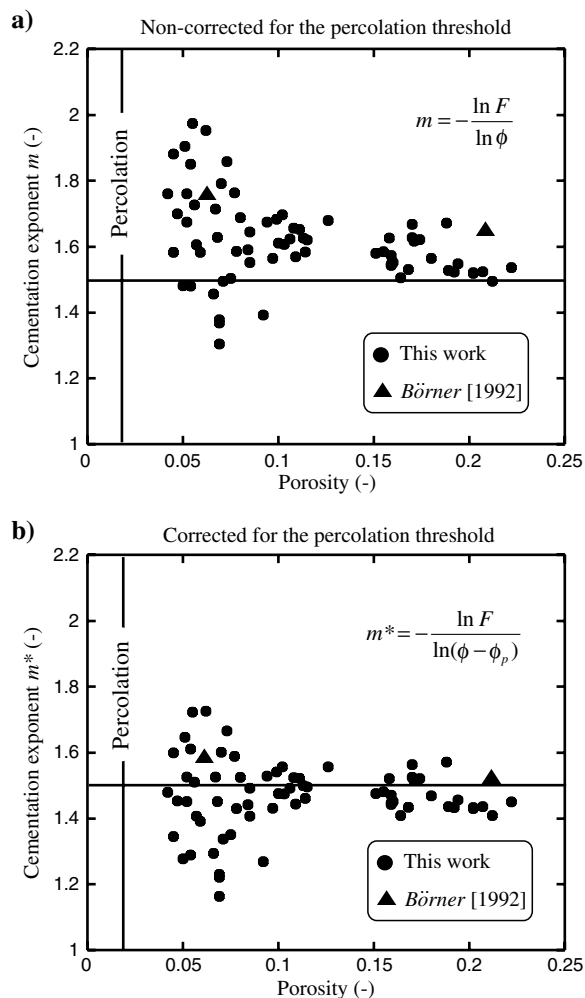


Figure 5. Cementation exponent versus connected porosity. When corrected with a percolation threshold, the cementation exponent has a value close to the theoretical value of 1.5 corresponding to spherical grains and obtained from the differential effective medium theory (Sen et al., 1981). Note the increase of the scatter in the data at low porosities, i.e., in the vicinity of the percolation threshold, probably because the size of the representative elementary volume becomes comparable (or is larger) to the size of the core samples. The data points from Börner (1992) concern the core samples F1 and F3.

behavior is surprising because permeability should be more sensitive to the effective porosity than to the total connected porosity (even for a sandstone with spherical grains, the effective porosity is expected to be different from the total connected porosity by a factor of 2 to 3). By effective porosity, we mean the porosity that is dynamically connected by the current lines. At the opposite, the connected porosity denotes the volume of connected voids over the volume of the rock. As discussed by [Revil and Cathles \(1999\)](#) and

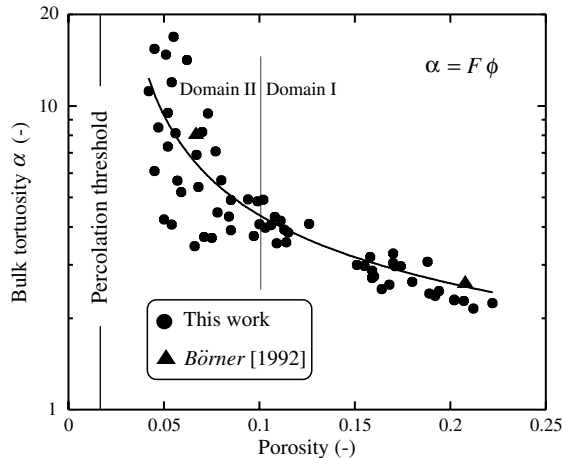


Figure 6. Bulk tortuosity of the pore space α versus connected porosity ϕ . Note the substantial increase in the bulk tortuosity of the pore space in the vicinity of the percolation threshold. The curve corresponds to the equation $\alpha = \phi(\phi - \phi_p)^{-m^*}$ with $m^* = 1.5$ (porosity exponent) and $\phi_p = 0.02$ (percolation porosity). The data points from [Börner \(1992\)](#) concern samples F1 and F3 of Fontainebleau sandstones.

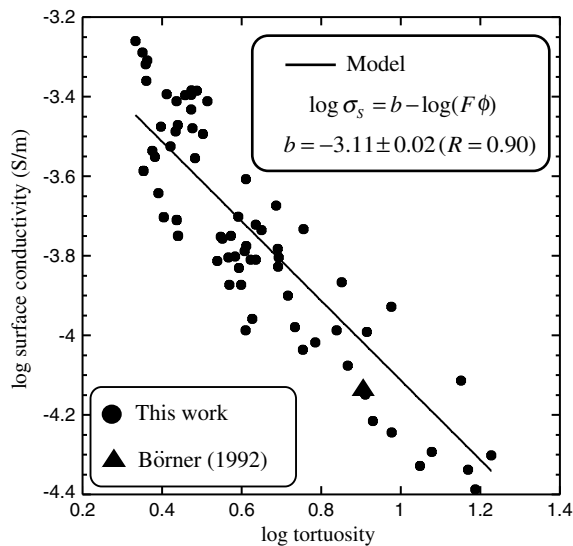


Figure 7. Surface conductivity versus bulk tortuosity of the pore space. The straight plain line corresponds to the best fit of the model equation with a single grain diameter. The data point from [Börner \(1992\)](#) concerns the sample F3 of Fontainebleau sandstone with a porosity of 0.068, a formation factor of $F = 117.7$ (tortuosity 8.00), a quadrature conductivity of $2 \times 10^{-6} \text{ S m}^{-1}$, a permeability of $138 \times 10^{-15} \text{ m}^2$, and a surface conductivity of $6.6 \times 10^{-5} \text{ S m}^{-1}$.

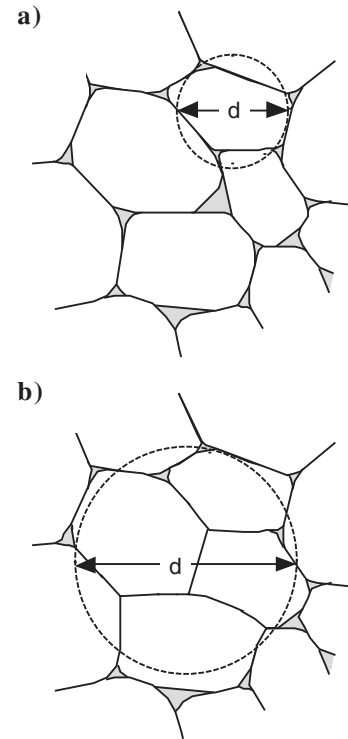


Figure 8. Sketch illustrating the potential effect of quartz cementation upon the effective grain diameter. (a) At high porosities, the effective grain diameter is equal to the size of the grains. (b) Some pore channels between grains are closed to fluids when the quartz sand undergoes cementation. There is an increase in the clustering of grains with cementation. However, this increase may not influence the local mean radius of curvature in the pores that remain open.

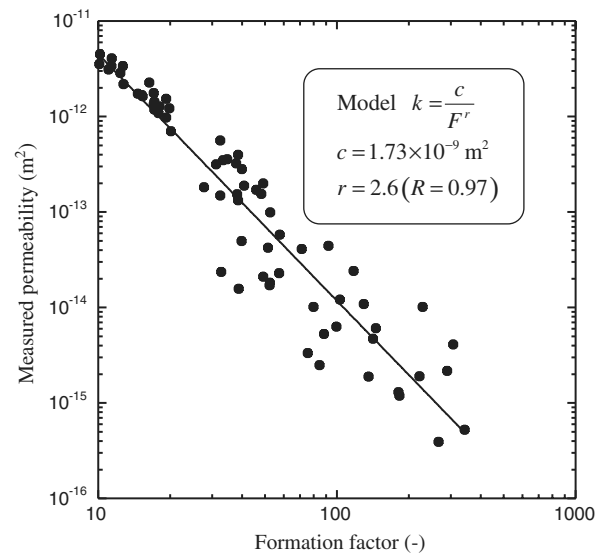


Figure 9. Permeability versus electrical formation factor. The formula derived in the main text is compared to experimental data measured using the full data set of the 69 Fontainebleau samples. The observed exponent for the dependence of permeability with the formation factor of $r = 2.6$ is close to the theoretical value $r = 3.0$.

Revil (2013b), the inverse of the formation factor is exactly an effective porosity. We have no explanation for this unexpected behavior.

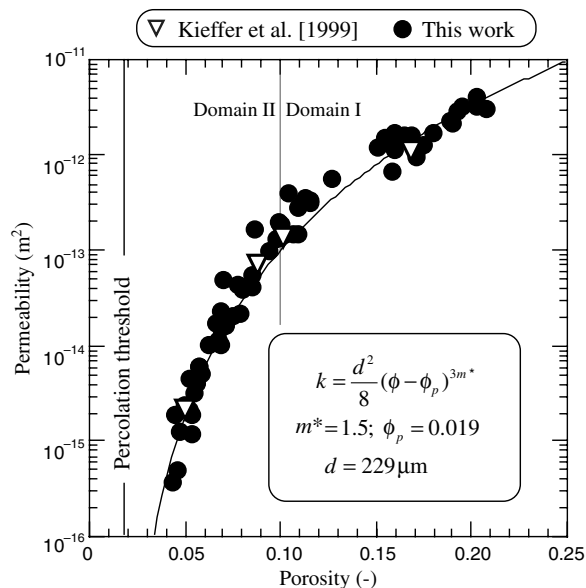


Figure 10. Measured permeability versus predicted porosity using the porosity, the percolation porosity, and the porosity exponent. The experimental data measured on 69 Fontainebleau samples were fitted with equation 12. Additional data are from Kieffer et al. (1999) for air permeability and total porosity. There is a good agreement between the model and the data. Note the strong decrease in permeability in the vicinity of percolation threshold.

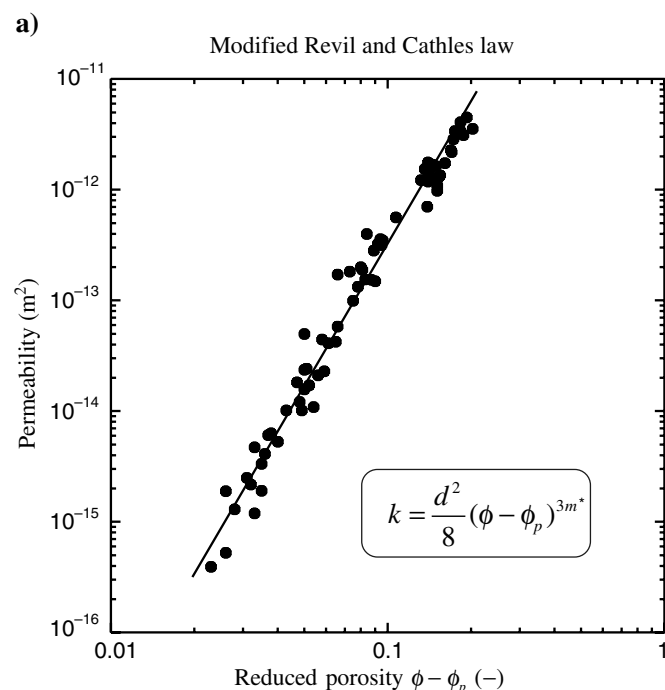


Figure 9 shows that permeability is inversely proportional to the formation factor at power 2.6, which is close to the theoretical value of 3 (see equation 11). Figure 10 provides a successful test of equation 12 using the theoretical value of the cementation exponent ($m = 1.5$), and the value of the percolation porosity determined from the relationship between formation factor and porosity (see Figure 4).

In Figure 11, we plot permeability and formation factor as a function of the reduced porosity (connected porosity minus the porosity at the percolation threshold). In a log-log plot, permeability plots linearly versus reduced porosity indicating that a Kozeny-Carman equation (even with a percolation threshold) is not a good choice to represent the permeability versus porosity data. In a log-log plot, the formation factor data follow a linear trend with reduced porosity indicating the ability of the modified Archie's law to capture the trend with a cementation exponent of 1.5 as predicted from differential effective medium (DEM) theory.

The next step is to examine whether we can infer permeability from surface conductivity and formation factor. From equation 8, the mean grain diameter can be replaced by the surface conductivity according to

$$d = \frac{1}{\sigma_s} \left(\frac{6Q_s \beta_{(+)} }{F\phi} \right). \quad (15)$$

Equations 11 and 15 yield

$$k \approx \frac{9Q_s^2 \beta_{(+)}^2}{2} \left(\frac{1}{\sigma_s^2} \right) \left(\frac{1}{F^5 \phi^2} \right). \quad (16)$$

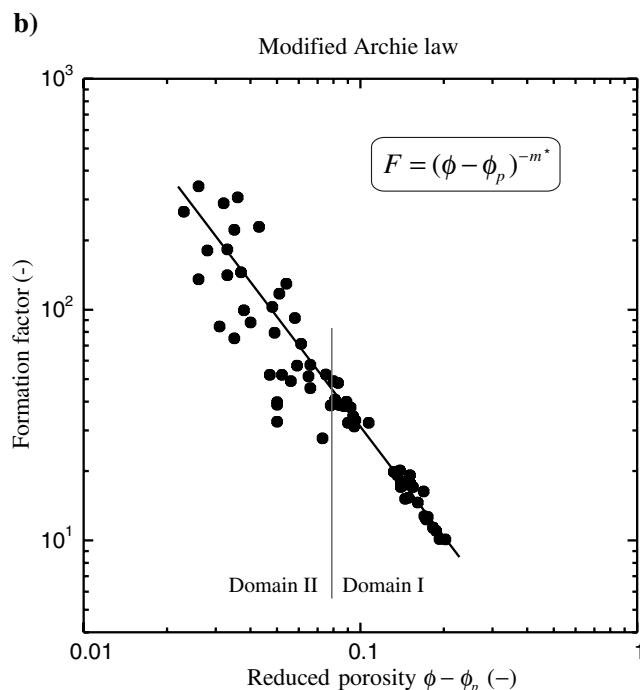


Figure 11. Permeability and formation factor as a function of the reduced porosity (connected porosity minus the porosity at the percolation threshold $\phi_p = 0.019$). We observe that once corrected for the percolation threshold, permeability and formation factor scale linearly, in a log-log plot, with reduced porosity. In both cases, we used $m^* = 1.5$. (a) Modified Revil and Cathles' (1999) equation ($R = 0.96$, $d = 282 \pm 40 \mu\text{m}$, and $a = 8$ kept fixed). (b) Modified Archie's equation (Archie, 1942, $R = 0.94$). Data source: this work.

The prediction of equation 16 is tested in Figure 12. We observe that the use of the formation factor porosity and the surface conductivity of the 69 rock samples can be used to predict the permeability with a coefficient of correlation $R = 0.83$. This is a fair prediction, but at the end, it is a bit disappointing for such simple porous material.

Resistivity index

The resistivity index (Ri) is defined as the ratio of the resistivity at a given saturation to the resistivity at 100% water saturation ($Ri \geq 1$). We first discuss the importance of surface conductivity in determining the Ri and the limit of the second Archie's equation. According to the model developed recently by Revil (2013a), the effect of saturation s_w on electrical conductivity is given by

$$\sigma = s_w^n \frac{1}{F} \sigma_w + s_w^{n-1} \frac{1}{F\phi} \lambda_s, \quad (17)$$

where λ_s appears as intrinsic surface conductivity defined by $\lambda_s = 6Q_s\beta_{(+)}^2/d$. In equation 17, n represents the second Archie exponent (saturation exponent). Equation 17 highlights the effects associated with the tortuosity of the pore space, the effects with excess tortuosity of the water phase associated with the partial saturation, and the influence of the two intrinsic conductivities λ_s and σ_w . The expression for Ri can now be written as

$$Ri = s_w^{-n} \left(\frac{\phi\sigma_w + \lambda_s}{\phi\sigma_w + \lambda_s/s_w} \right). \quad (18)$$

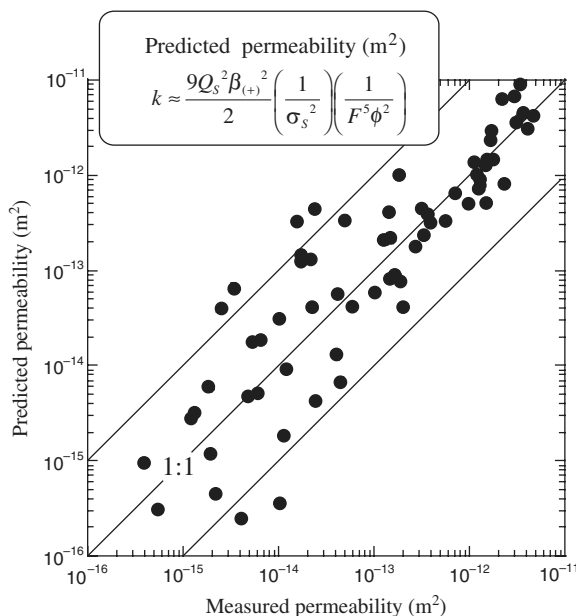


Figure 12. Predicted versus measured permeabilities for the 69 samples. The predicted permeability is determined from the formation factor, the porosity, and the surface conductivity. We used the following values of the parameters: $Q_s = 0.64 \text{ C m}^{-2}$ and $\beta_{(+)}(\text{Na}^+, 25^\circ\text{C}) = 5.2 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$. Note that there are no fudge factors in the model used here.

It follows that the condition for which surface conductivity can be neglected is for salinities and saturation such as $\sigma_w \gg \lambda_s/\theta$, where $\theta = s_w\phi$ denotes the volumetric water content. In our case, using $Q_s = 0.64 \text{ C m}^{-2}$, $\beta_{(+)}(\text{Na}^+, 25^\circ\text{C}) = 5.2 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, and $d = 229 \text{ }\mu\text{m}$ yields $\lambda_s = 8.72 \times 10^{-4} \text{ S m}^{-1}$. For clean sandstones, the Ri is usually measured at such high salinity and equal to $Ri = s_w^{-n}$ based on the second Archie's equation neglecting surface conductivity.

Yanici et al. (2013) observe that the Ri data for the Fontainebleau sandstone does not conform to Archie's equation. They measured the Ri at a high salinity (NaCl, 2% by weight NaCl brine) for a Fontainebleau sandstone with 0.159 porosity (15.9%) (Figure 13). We observe that the data obey Archie's equation down to a critical saturation. We postulate that this critical saturation is related to the percolation porosity by $s_r = \phi_p/\phi$, which is approximately equal to 0.12 in the case of the data shown in Figure 13, not far from the observed value of 0.15.

INDUCED POLARIZATION

Because of the electrical double layer coating the surface of the grains, the Fontainebleau sandstone is expected to exhibit some degree of polarization at low frequencies. This polarization is related to the reversible storage of electrical charges in an alternating current or electrical field. To account for the phase shift between the electrical current and the electrical field, the conductivity is often written as a complex number $\sigma^* = \sigma' + i\sigma''$ in which σ' and σ'' denotes the in-phase and quadrature conductivity, respectively, and i denotes the pure imaginary number. The in-phase conductivity characterizes the conductive property of the material and is basically the one investigated in the previous sections. The quadrature

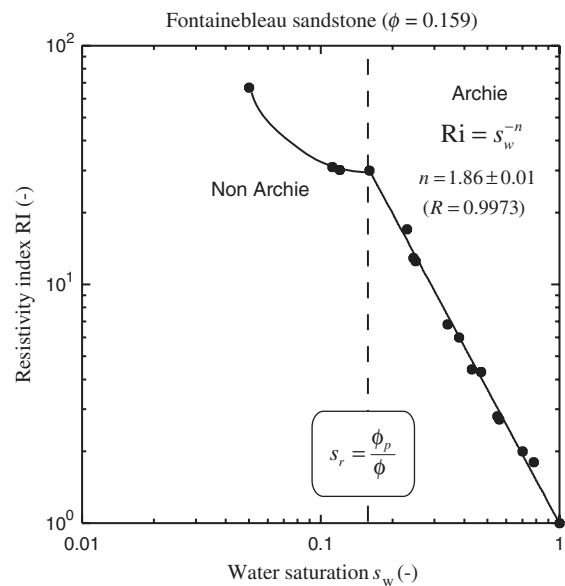


Figure 13. Ri of Fontainebleau sandstone versus water saturation at high salinity (negligible surface conductivity). The data exhibit an Archie-type behavior with the second Archie's exponent $n = 1.86 \pm 0.01$ down to a residual water saturation approximately given by the ratio of the porosity at percolation (0.019) and the measured porosity (0.159). Data are from Durand (2003), Knackstedt et al. (2007), Han et al. (2009) (sample porosity 0.22), and Yanici et al. (2013).

conductivity characterizes the polarization of the material. The ability of porous rocks to store reversibly electrical charges at low frequencies is called *induced polarization* in geophysics. Preliminary measurements of induced polarization of the Fontainebleau sandstone are reported below.

Theory

Revil (2012, 2013a) advance a relationship between surface conductivity and quadrature conductivity determined from spectral-induced polarization measurements. Revil (2013b) uses two dimensionless numbers to express the complex electrical conductivity in the following form:

$$\sigma^* = \frac{1}{F} \sigma_w [1 + \text{Du}(1 - iR)], \quad (19)$$

where the two dimensionless numbers are defined as

$$\text{Du} = \frac{F\sigma_s}{\sigma_w} = \left\{ \frac{\Theta\rho_s[\beta_{(+)}(1-f) + \beta_{(+)}^S f]}{\phi\sigma_w} \right\} \text{CEC} \quad (20)$$

and

$$R \equiv -\frac{\sigma''}{\sigma_s} = \frac{\beta_{(+)}^S f}{[\beta_{(+)}(1-f) + \beta_{(+)}^S f]}. \quad (21)$$

The dimensionless number Du is called the *Dukhin number*, and it represents the ratio of surface (σ_s) to bulk conductivity (σ_w/F) (Dukhin and Shilov, 1974, 2002) whereas the dimensionless number R corresponds to the ratio between quadrature conductivity (σ'') and surface conductivity (σ_s) and was introduced by Revil (2013b). The quadrature conductivity is given explicitly by

$$\sigma'' \approx -\left(\frac{1}{F\phi}\right)\rho_s f \beta_{(+)}^S \text{CEC} \quad (22)$$

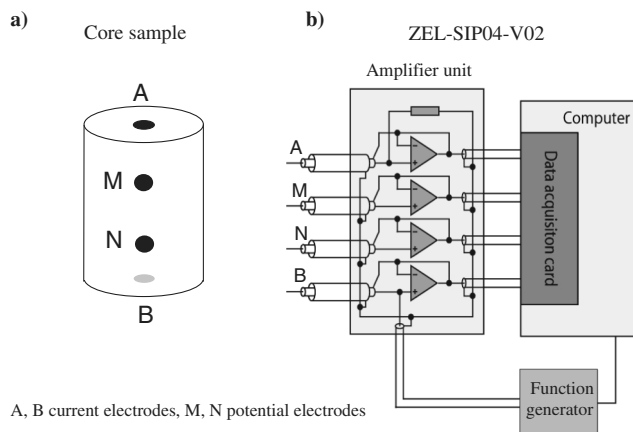


Figure 14. Experimental setup. (a) Position of the electrodes on the core sample. (b) Position of the current (A and B) and voltage (M and N) electrodes. The data acquisition system operates in the frequency range from 1 to 45 kHz with a phase accuracy close to 0.1 mrad below 1 kHz.

whereas from equation 6, the surface conductivity is given by

$$\sigma_s \approx \left(\frac{1}{F\phi}\right)\rho_s[\beta_{(+)}(1-f) + \beta_{(+)}^S f] \text{CEC}. \quad (23)$$

For clean sands, as explained by Revil (2013a, 2013b), the mobility of the Stern layer is the same as in the bulk pore water. It follows that the Du number is given by $\text{Du} = 6Q_s\beta_{(+)} / (d\sigma_w\phi)$. However, Fontainebleau sandstone cannot, possibly, be considered as a clean sandstone in terms of interfacial properties. The incorporation of impurities such as Al and Fe ions into the crystalline framework of silica can substantially modify the electrical properties of the mineral surface (Iler, 1979). This point will be discussed further at the end of this paper.

DATA ACQUISITION

The acquisition of the data was performed with the ZEL-SIP04-V02 impedance meter built by Egon Zimmermann (Figure 14; see

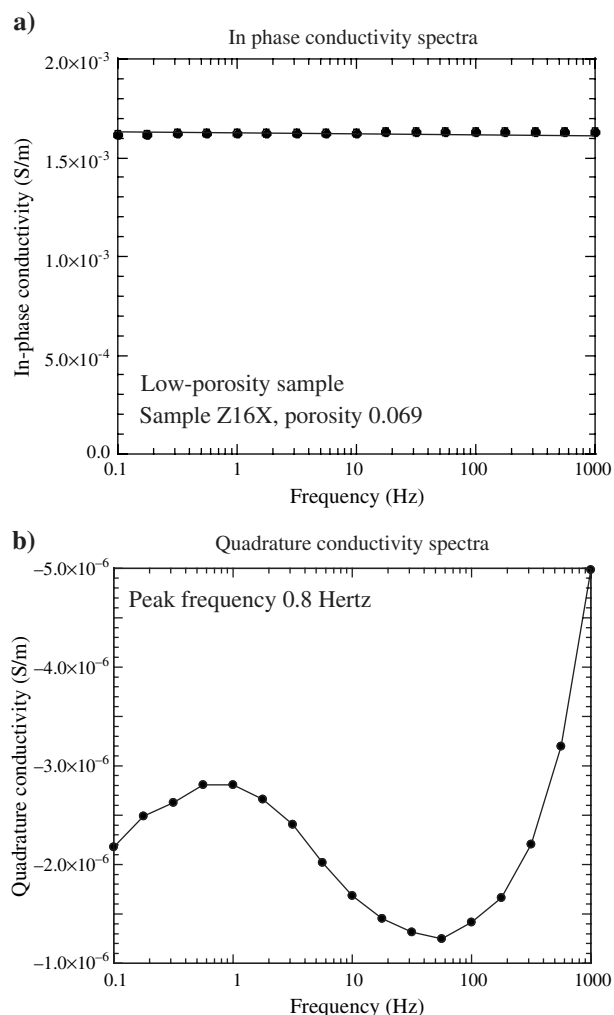


Figure 15. Conductivity spectra for sample Z16X. (a) In-phase conductivity showing nearly no dependence with frequency. (b) Quadrature conductivity showing a relaxation peak at 0.8 Hz. The high-frequency polarization is probably due to Maxwell-Wagner polarization.

also Zimmermann et al., 2008a, 2008b). In-phase and quadrature conductivities of sample Z16X are shown in Figure 15. The data indicate that the in-phase conductivity is fairly frequency independent and in agreement with the conductivity performed with the two-electrode system. Figure 15b shows the quadrature conductivity versus frequency curve for one sample. In this preliminary investigation, we are only interested by the quadrature conductivity at a single frequency. We choose 0.1 Hz as having a good signal-to-noise ratio for the phase, and the quadrature conductivities will be reported at this frequency.

Comparison between data and theory

In Figure 16, we compare the relaxation times determined from the low-frequency polarization and the pore size either determined from the permeability (Figure 16a) or from the mercury porosimetry measurements (Figure 16b). In Figure 16b, using a combination of clayey sandstones and two Fontainebleau sandstones, we observe that the relaxation time is correlated with the peak pore size that can be determined from mercury porosimetry measurements shown in Figure 16c. The fact that the Fontainebleau data agrees with the trend offered by clayey materials (see Revil, 2013b) seems to indicate that the surface properties of the Fontainebleau sandstone are not those of clean sands.

The next characteristic to be tested concerns the relationship between quadrature conductivity and bulk tortuosity. As shown in Figure 17, quadrature conductivity appears to be inversely proportional to bulk tortuosity of the pore space computed as the product between formation factor and porosity. Indeed, from equation 22, we can write the quadrature conductivity as $\log \sigma'' \approx c - \log(F\phi)$ with $c = \log[\rho_s f \beta_{(+)}^s \text{CEC}]$. In Figure 17, we obtain $c = -5.22$ by fitting the data with this equation. We will now determine the theoretical value of this parameter.

We first investigate the relationship between surface conductivity and quadrature conductivity. In Figure 18, we plot our data together with the Fontainebleau data point of Börner (1992) and the data set discussed by Weller et al. (2013) for clayey materials only. Our data set seems in agreement with the linear trend discussed by Weller et al. (2013). We can investigate this relationship a bit further. If we assume that the grains have the properties of pure silica in terms of electrochemical properties, we can use the model of Leroy et al. (2008) to compute the fraction of counterions located in the Stern layer f . The results are shown in Figure 19 for various values of the pH and salinity of NaCl solutions. For the pH of water in equilibrium with the atmosphere and for the salinity used in this work, the value of f is small (on the order of 0.03). This indicates that the conditions used in this work, the fraction of counterions in the Stern layer is small, and therefore the quadrature conductivity is in turn small. If we take a mobility of the counterions

equal to the mobility of the same ions in water ($\beta_{(+)}(\text{Na}^+, 25^\circ\text{C}) = 5.2 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$), $f = 0.03$, $\text{CEC} = 5.80 \text{ C kg}^{-3}$, we obtain $c = \log[\rho_s f \beta_{(+)}^s \text{CEC}] = -4.6$ not too far from the value $c = -5.2$ obtained in Figure 17 in fitting the data. According to Figure 19, much higher quadrature conductivity values are expected if the pH is increased to 9 as discussed by Leroy et al. (2008).

DISCUSSION

There are four points deserving additional discussion in the present paper. The first one is related to the importance of surface conductivity in clean sands. The second one is related to the observation and the model prediction that surface conductivity is controlled by bulk tortuosity of pore space and not, as usually (and maybe naively) thought by the tortuosity of the surface of the grains. The third point is related to the grain or pore size control of the surface properties, and finally, the last point concerns the effect

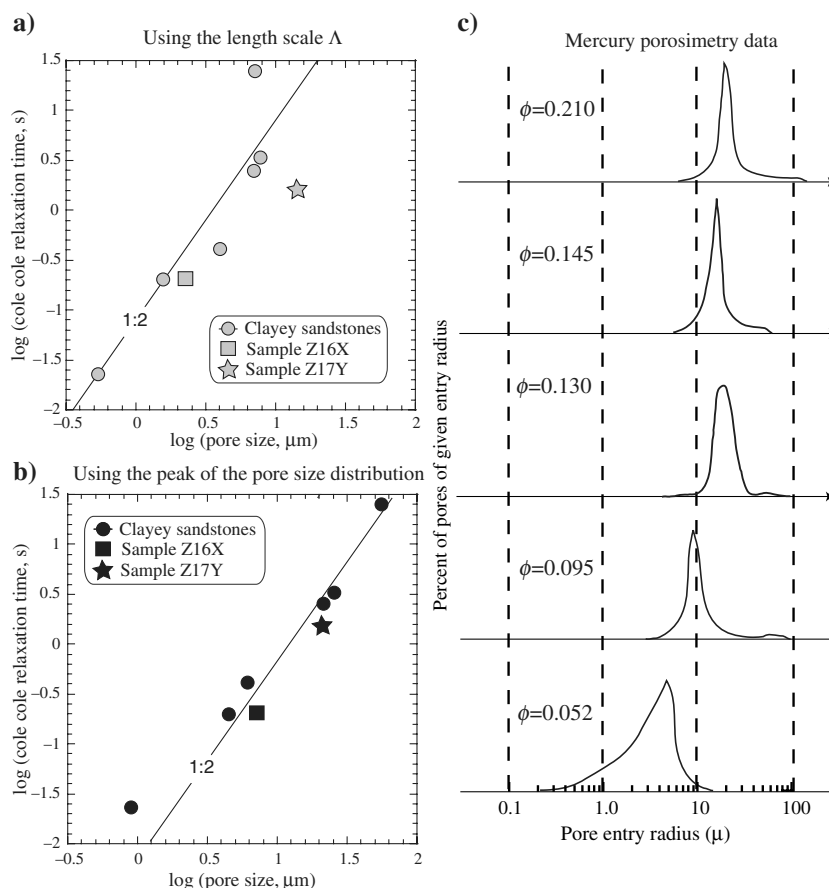


Figure 16. Relationship between the Cole–Cole relaxation time and two different pore sizes reported in Revil et al. (2014) submitted for a collection of clayey sandstone samples. (a) Correlation between the Cole–Cole relaxation time and the pore size $\Lambda = (8kF)^{1/2}$ determined from the formation factor and permeability (2.2 μm for sample Z16X and 14.9 μm for sample Z7Y). (b) Correlation between the Cole–Cole relaxation time (or the characteristic time associated with the peak frequency for the Fontainebleau sandstones) and the peak of the pore size distribution determined from mercury intrusion measurements. (c) For the Fontainebleau sandstone, we use the mercury porosimetry data of Bourbié and Zinsner (1985) based on the porosity value of the core sample (7 μm for sample Z16X and 20 μm for sample Z7Y). The data for the sample at 13% porosity are from David et al. (2013).

of impurities (Al, Fe) regarding the electrochemical properties on the surface of the grains.

Importance of surface conductivity in clean sandstones

Surface conductivities of Fontainebleau sandstone are in the range $0.5 \times 10^{-4} \text{ S m}^{-1}$ – $5 \times 10^{-4} \text{ S m}^{-1}$ at 20°C . These values are not negligible, and formation factors or R_i interpreted under the assumption that the surface conductivity is negligible are likely

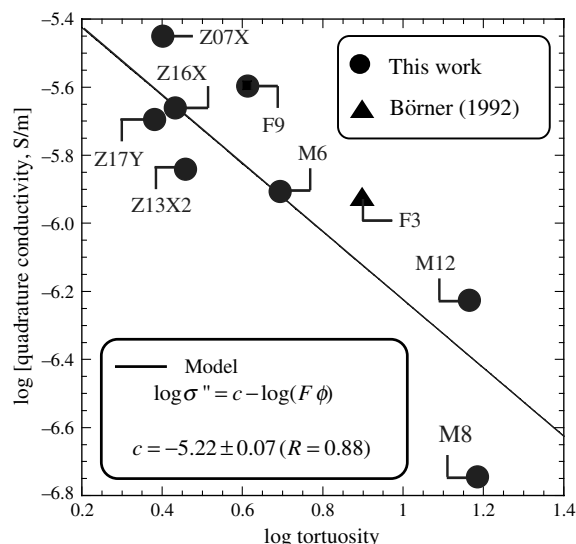


Figure 17. Quadrature conductivity versus bulk tortuosity of the pore space. The straight line corresponds to the best fit of the model equation. The data point from Börner (1992) concerns the sample F3 of Fontainebleau sandstone with a porosity of 0.068, a formation factor of $F = 115.3$ (tortuosity 7.84), a quadrature conductivity of $2 \times 10^{-6} \text{ S m}^{-1}$, and a surface conductivity of $7 \times 10 \text{ S m}^{-1}$.

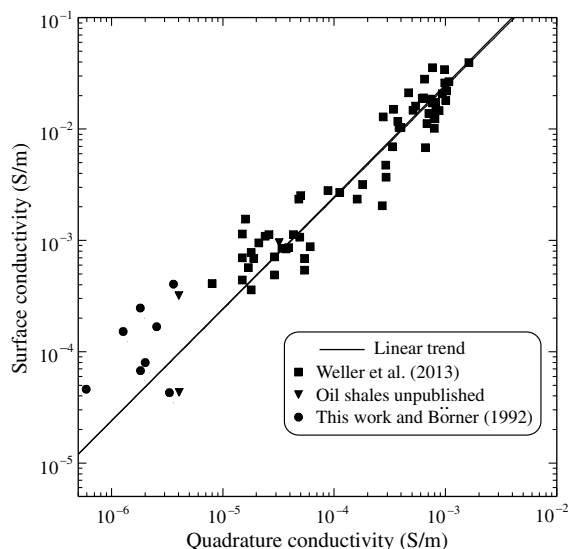


Figure 18. Comparison of the relationship determined by Weller et al. (2013) (only the clayey materials) and additional data including oil/gas shales with low porosities and samples of Fontainebleau sandstones (seven samples from this work and sample F3 from Börner, 1992).

unreliable. For instance, Gomez et al. (2010) did not account for surface conductivity and obtained formation factors and porosity exponents that were inconsistent with the present database (their porosity exponent, approximately 1.8, is too high for spherical grains). It follows that the formation factor determined by neglecting surface conductivity in freshwater environments can be quite different from

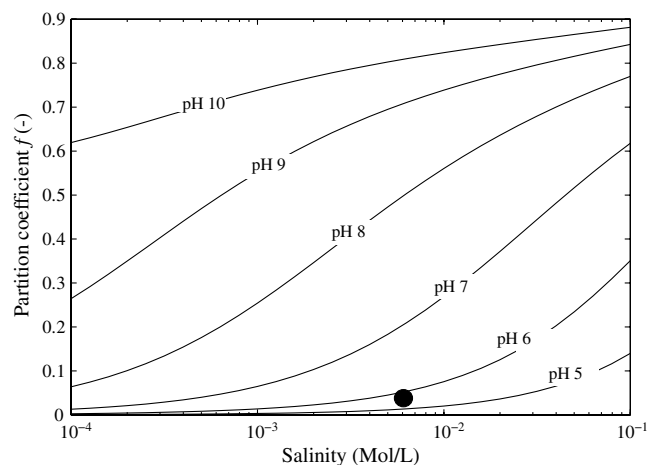


Figure 19. Determination of the partition coefficient through a triple layer model for silica (Leroy et al., 2008) for different values of the pH and salinity of NaCl solutions. The filled circle represents the conditions used for this work in terms of salinity and pH (5.6).

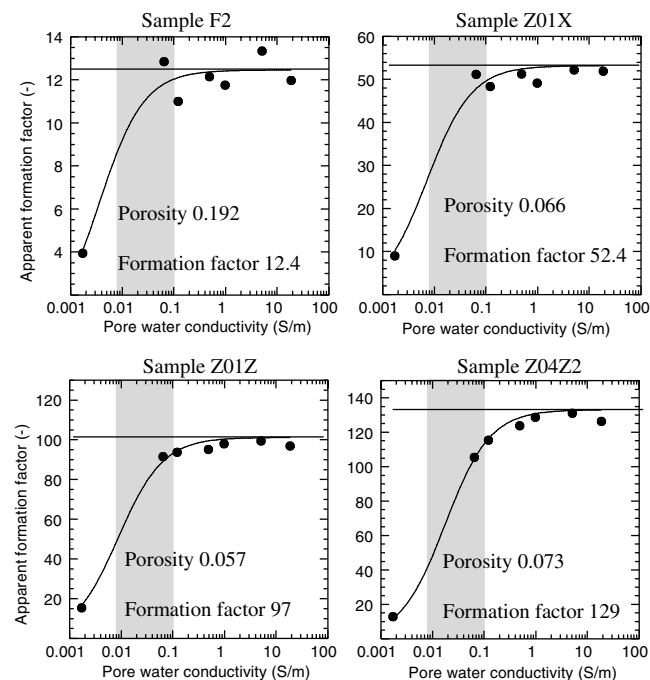


Figure 20. Relationship between apparent formation factors (defined as the ratio of the resistivity by the resistivity of the pore water) and intrinsic formation factors accounting for the effect of the surface conductivity in the relationship between the conductivities of the sample and the pore water. The horizontal line identifies the value of the intrinsic formation factor F , whereas the curved line identifies a fit with equation 4 in terms of an apparent formation factor. The gray areas correspond to typical conductivity values of freshwater aquifers ($80\text{--}1000 \mu\text{m cm}^{-1}$).

the intrinsic formation factor of core samples. In Figure 20, we show the apparent formation factor defined by $F_a = \sigma_w / \sigma$ as a function of the intrinsic formation factor F for four samples from this work. The same analysis is shown in Figure 21 for samples F3 from Börner (1992) and Z14Y (this work). These two samples have roughly the same porosity, and therefore, the comparison also indicates the consistency between our measurements and the measurements from Börner (1992) for sample F3. In both cases, we can conclude that in freshwater environments (pore water conductivity in the range 80–1000 $\mu\text{m cm}^{-1}$), there is a substantial difference between the apparent and intrinsic formation factors indicating that the surface conductivity is important to interpret field data.

Because the surface and quadrature conductivities are associated with the electrical double layer, there are also sensitive to the chemical composition of the bulk pore water. Vaudelet et al. (2011a, 2011b) and Revil et al. (2013b) show how the composition of the pore water can have an impact on the quadrature and surface conductivities. Clearly, additional work is needed to correct for the compositional effect of pore water or to take advantage of it in tracer studies.

Why does the bulk tortuosity of the pore space control surface and quadrature conductivities?

A somewhat naive approach of the conductivity problem of porous media is to assign a distinct tortuosity to bulk and surface conduction processes with a higher tortuosity for surface conduction. This approach, however, is not reliable. As discussed in detail by Johnson et al. (1986), Bernabé and Revil (1995), and Revil and Glover (1997), in the high-salinity limit (surface conductivity is a perturbation of the bulk electrical conductivity), the electrical field in the pore space is controlled by the pore network, not by the distribution of surface conductances. It follows that the bulk tortuosity

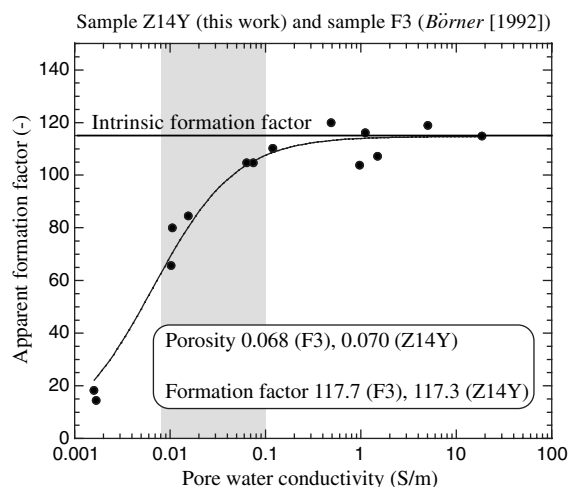


Figure 21. Relationship between apparent formation factors (defined as the ratio of the resistivity by the resistivity of the pore water, horizontal line) and intrinsic formation factors accounting for the effect of the surface conductivity in the relationship between conductivity of the sample and the pore water conductivity (curved line). We use samples F3 from Börner (1992) and Z14Y (this work). The data for these two samples characterized by the same porosity and formation factor are consistent. The surface conductivity is estimated to be $(5.8 \pm 0.7) \times 10^{-5} \text{ S m}^{-1}$. The gray area corresponds to the typical conductivity values of freshwater aquifers (80–1000 $\mu\text{m cm}^{-1}$).

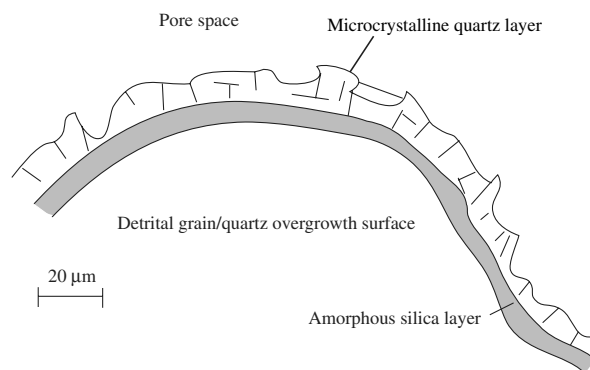


Figure 22. Microstructure of the grains in the Fontainebleau sandstone. The detrital grains with minor amounts of quartz overgrowth (the Fontainebleau sandstone has undergone a minor degree of early quartz cementation) are coated by a layer of amorphous silica. In turn, this layer is coated with a microcrystalline quartz layer.

of the pore space, not the tortuosity along the surface of the minerals, controls surface conductivity in the high-salinity limit. As discussed by Revil and Glover (1997), the situation would be different at low salinities for which the conductances along the surface of the grains would dominate the rock's electrical conductivity. However, this case has no practical applications because it would occur at pore water mineralization below the lower conductivity of freshwater aquifers. To our knowledge, this is the first time that the dependence of surface conductivity (Figure 7) and quadrature conductivity (Figure 17) with bulk tortuosity of the pore space is shown through experimental data.

Grain-size- or pore-size-based models

When we are dealing with a granular material, we usually write the transport properties in terms of mean grain size or grain-size distribution (see, for instance, Revil and Florsch, 2010). Conversely, for low-porosity media, we write the transport properties in terms of the pore size or the pore size distribution. One question may arise for the Fontainebleau sandstone: which formulation should be chosen? Indeed, in the case of the Fontainebleau sandstone, the porosity covers a broad range of values. In our approach above, we have chosen a grain-based model. It will be interesting in the future to examine magnetic resonance and mercury porosimetry data, and spectral-induced polarization to elucidate a causal relationship between pore and grain sizes. We can possibly consider a high-porosity domain (domain I) in which the control on the transport properties is made by the grain-size distribution and a low-porosity domain controlled by the pore size distribution. Looking at the permeability/porosity and the formation factor/porosity curves, it seems that the transition between domain I and domain II is around a porosity of 0.10 (see, for instance, Figures 6, 10, and 11). This assumption will need to be explored further in future contributions.

Impurities on the surface of the silica grains

Another question is to know if the electrochemical properties of the surface of the grains of the Fontainebleau sandstones are those of pure silica or if a small amount of impurities (aluminum and iron) can play a strong role. First, amorphous silica coats the surface of many detrital grains in the Fontainebleau formation (French and

Worden, 2013). This amorphous silica is in turn coated by microcrystalline silica (Figure 22). We know that alumina is easily incorporated into the crystalline framework of silica (e.g., Chappex and Scrivener, 2012). The incorporation of this alumina into the crystalline framework in the vicinity of the mineral surface may change its electrochemical properties. Further works (for instance, zeta potential measurements) will be needed to elucidate the surface speciation model of the surface of the Fontainebleau sandstone.

CONCLUDING STATEMENTS

The following conclusions have been reached for the Fontainebleau sandstone, a clean (clay-free) sandstone:

- 1) Even a clean sandstone such as the Fontainebleau sandstone exhibits surface conductivity along the grain surface. We have shown that for freshwater aquifers (conductivity in the range 80 to 1000 $\mu\text{m}/\text{cm}$), this surface conductivity should be explicitly accounted for when interpreting electrical resistivity measurements. In other words, the difference between the apparent and intrinsic formation factors is substantial. It can reach one order of magnitude for a pore water conductivity of 0.001 S m^{-1} for the low-porosity core samples.
- 2) The surface conductivity of Fontainebleau sandstone is controlled by the tortuosity of the pore space as predicted by the recent model of Revil but not by previous electrical conductivity models such as the popular Waxman and Smits' model. Once again, this behavior indicates that great care should be exercised when using petrophysical models to predict or interpret surface conductivity along the surface of grains.
- 3) Electrical conductivity and permeability data exhibit a percolation threshold of about 2% porosity. Introducing this percolation threshold in the relationship between formation factor and porosity improves the prediction of Archie's equation. This conclusion was reached for the relationship between the permeability and the porosity. The prediction model of power-law model of Revil and Cathles to predict permeability can be improved with the introduction of the same percolation porosity used in Archie's equation.
- 4) Quadrature conductivity is controlled by the bulk tortuosity of the pore space. The surface and quadrature conductivity seem correlated in a way that is consistent with clayey materials. This behavior may be due to the existence of impurities on the surface of silica including the presence of small amounts of iron and alumina in the crystalline framework.

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